

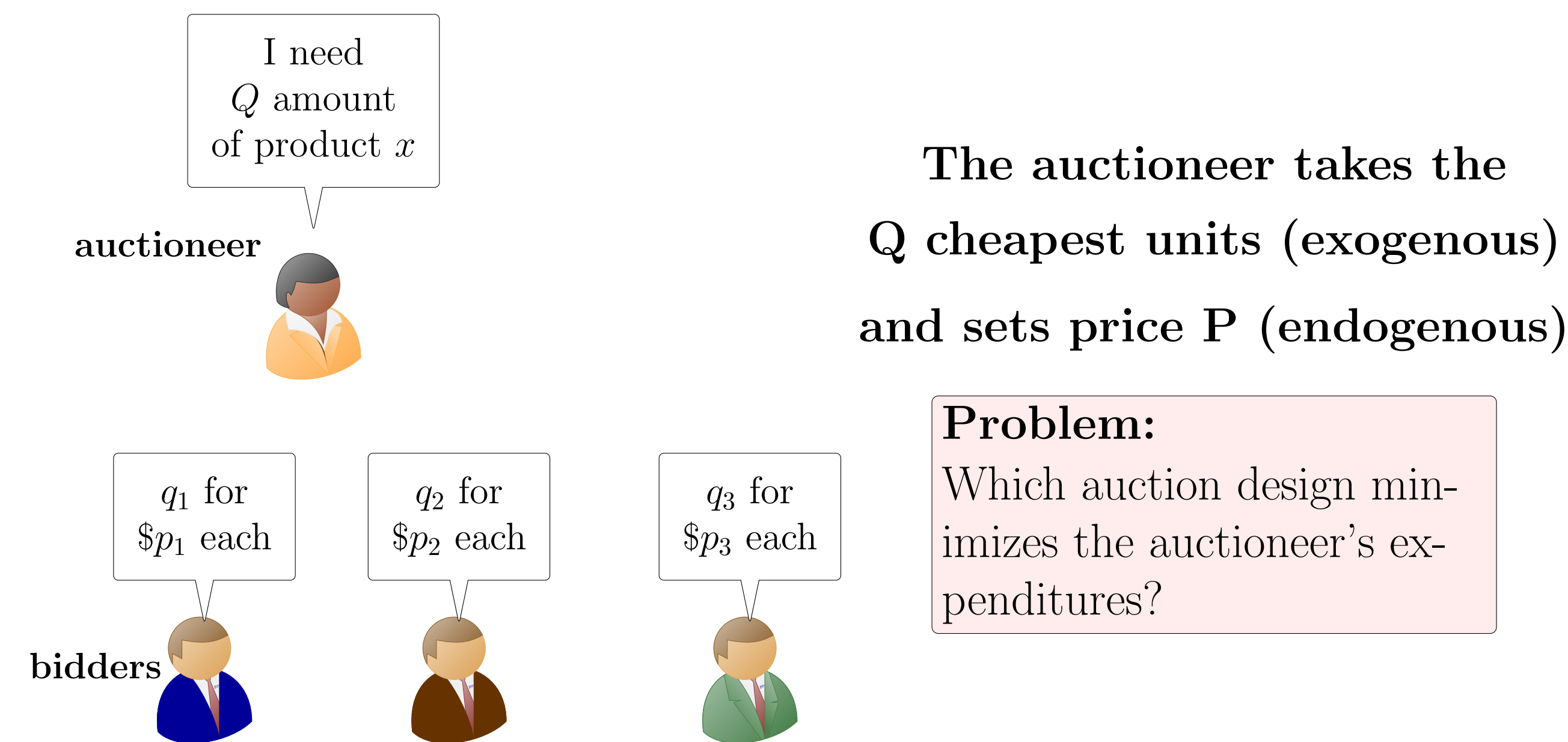
Neural Network Estimation of Counterfactual Bayesian Nash Equilibria in Auction Games

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20-Second Core Idea

- Learn Nash equilibria in Bayesian games numerically.
- Bid strategies are neural networks trained via pseudogradient ascent.
- Equilibria emerge from repeated self-play updates.

Reverse Auctions



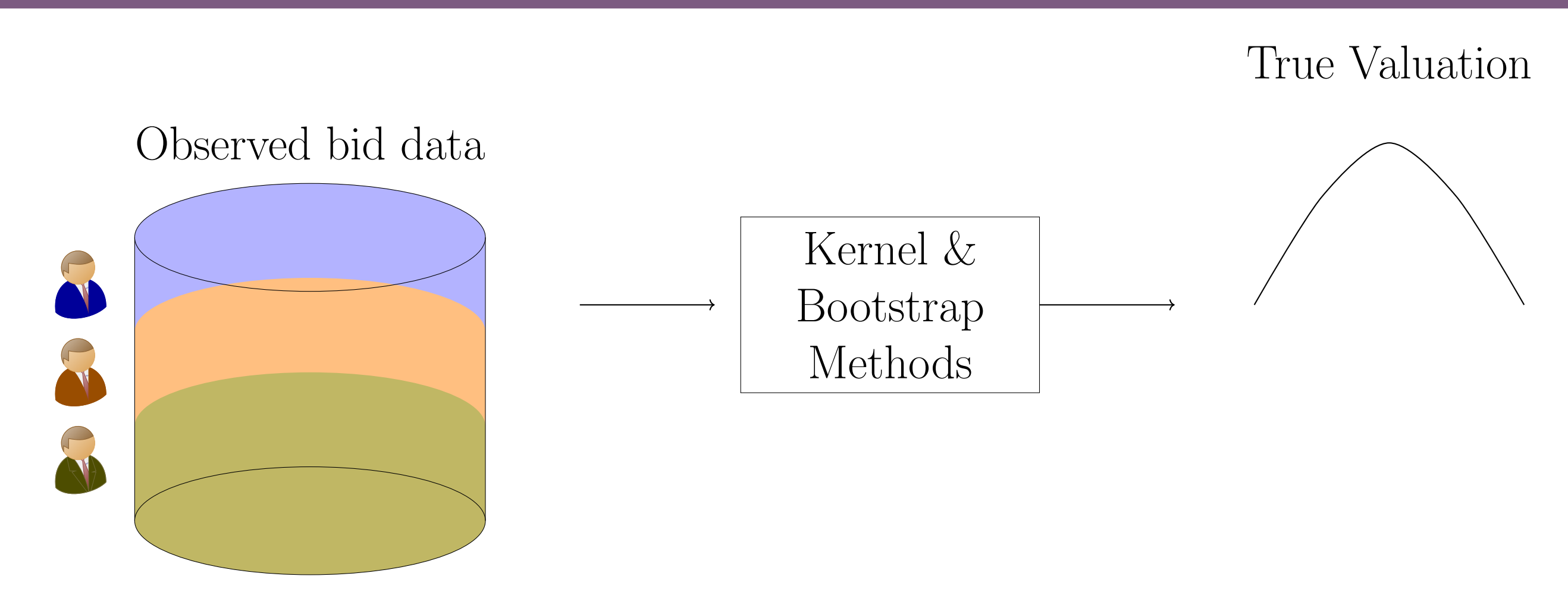
- Compare expenditures across different auction designs (uniform-price vs. discriminatory, etc.).
- Counterfactual BNEs required, but analytically intractable.
- Relevant in electricity markets, spectrum, procurement, and many others.

A Neural Network Approach

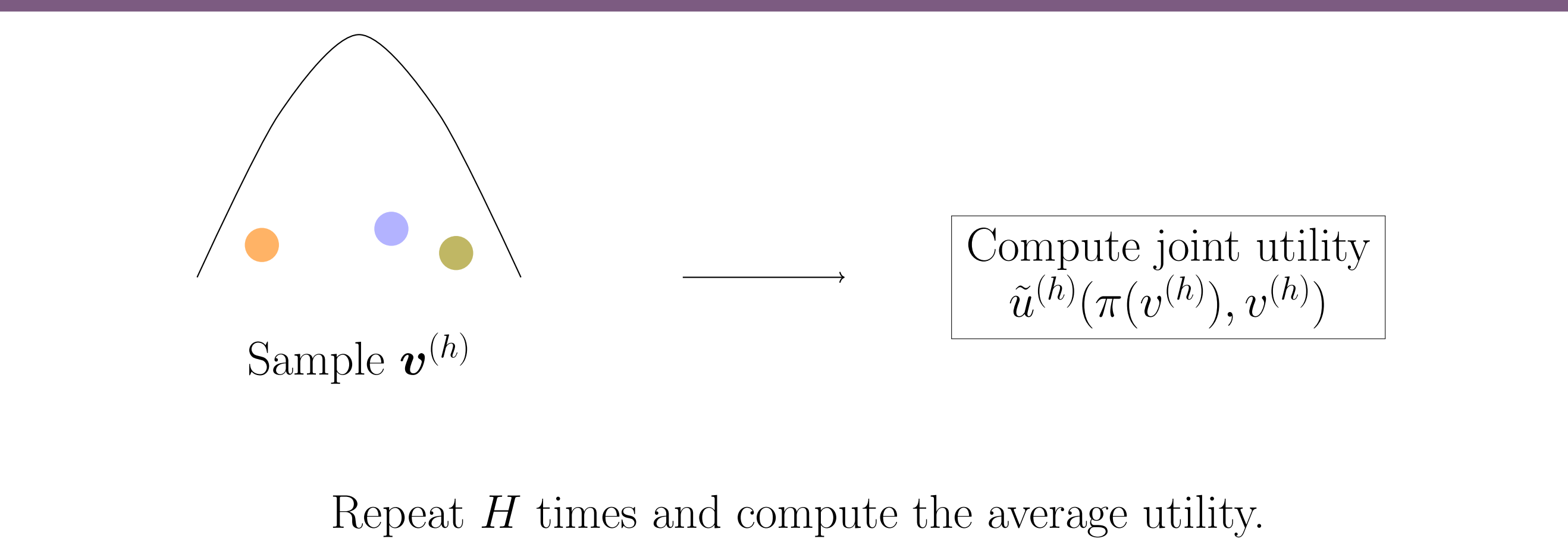
	Traditional Economics Bayesian Game	Neural Networks Proxy Game
Strategy	$b_i(v_i)$	$\pi_i(v_i; \theta_i)$
Target Function	$u(b_i(v_i), v_i)$	$u(\pi_i(v_i; \theta_i), v_i)$
Limitations	Intractable system of differential equations	Backpropagation fails; requires evolutionary updates

Notation: b_i : bid, v_i : valuation, q_i : quantity,
 u_i : utility (discontinuous, $q_i(b_i - v_i)$ if win, 0 otherwise)

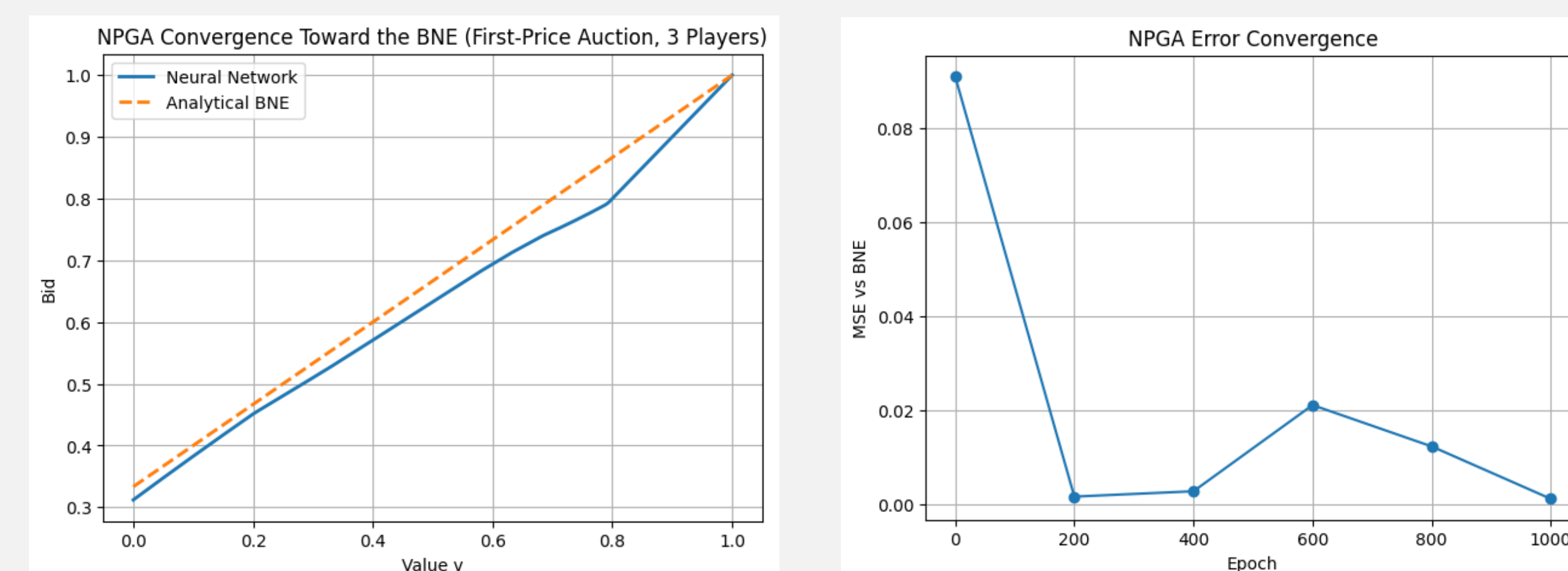
Step 0: Recover Valuation Functions



Step 1: Sampling



Convergence: High-quality Neural Network fit



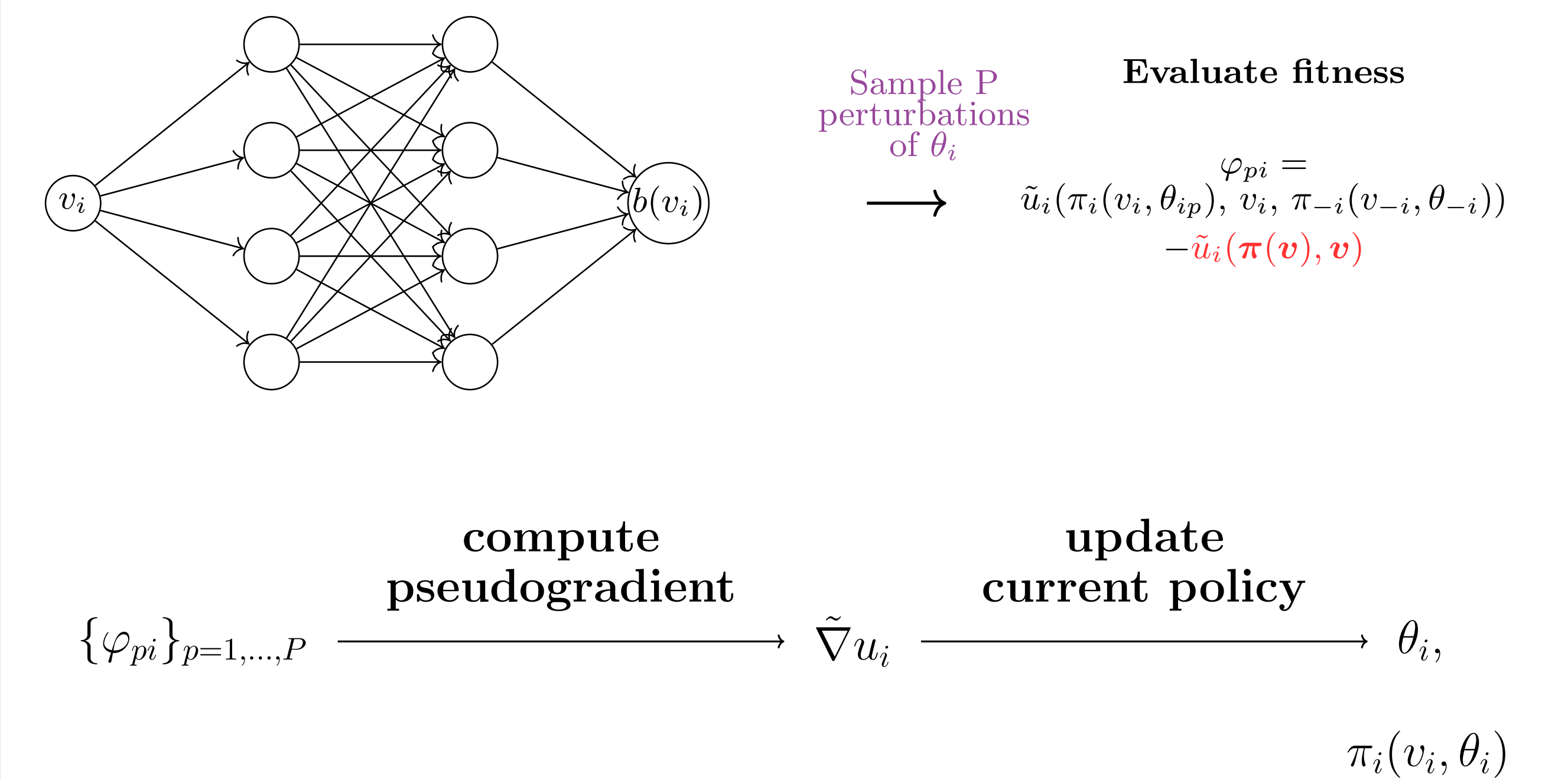
Optimal bids for 3 players in a single-item auction with uniform valuations.

Policy Network Architecture

- Fully connected neural networks with two hidden layers (10 units each).
- SeLU activation in hidden layers.
- ReLU (or sigmoid) activation in the output layer.

Step 2: Training

For each $i \in I$, agent i 's current policy is given by the network $\pi_i(v_i, \theta_i)$.



Step 3: Iterating

For $t = 1, 2, \dots$ repeat Steps 1 and 2 until convergence:

$$\|\tilde{\mathbf{u}}^t - \tilde{\mathbf{u}}^{t-1}\| < \varepsilon$$

Obtaining a ε -BNE

Final Remarks

- Neural networks provide an effective tool for learning counterfactual equilibria in reverse auctions.
- Auction design decisively shapes strategic bidding and procurement costs, underscoring the value of identifying cost-efficient mechanisms.
- Future work: application to energy auctions, where this framework can guide mechanism selection and indirectly influence household electricity costs.

References

- [1] M. Bichler, M. Fichtl, S. Heidekrüger, et al., «Learning equilibria in symmetric auction games using artificial neural networks», *Nature Machine Intelligence*, vol. 3, pp. 687–695, 2021. DOI: 10.1038/s42256-021-00365-4. [Online]. Available: <https://doi.org/10.1038/s42256-021-00365-4>.